Raunak Kumar

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WORK EXPERIENCE

01/10/2025 - CURRENT - BANGALORE, INDIA

FOUNDING AI RESEARCH INTERN MANDRAKE BIO

- Developing a **unified Bio-LLM** on a **Gemini-4B backbone** for joint protein **sequence and function modelling** enabling aligned cross-modal biological representations.
- Implemented **CLM-driven vocabulary expansion** with **LoRA adapters** to encode richer biochemical semantics while preserving pretrained knowledge.
- Worked with **RFdiffusion-based generative frameworks** for protein sequence–structure modelling and **denovo enzyme design.**
- Engineered **scalable HPC training, distributed fine-tuning, and benchmarking workflows** for large, heterogeneous biological datasets.

15/05/2025 - CURRENT - ROORKEE, INDIA

UNDERGRADUATE RESEARCHER PROF. JITIN SINGLA

Template-free Subtomogram Alignment & Classification:

- Developed and implemented a new template-free alignment method for **cryo-ET subtomograms**
- Engineered distributed subtomogram alignment routines that **reduced runtime by 20× and enabled 4× larger batch processing** achieving improved rotational refinement compared to prior methods even on pre-aligned subtomograms
- Investigating clustering strategies for structural class discovery and preparing a manuscript on the method for submission to the **Journal of Structural Biology**

01/01/2025 - 04/05/2025 - ROORKEE, INDIA

UNDERGRADUATE RESEARCHER COMMUNICATION SYSTEMS LAB

Cognitive Workload Classification using **TDA**:

- Used EEG recordings from the WAUC dataset to study cognitive workload levels.
- Extracted **persistent-homology-based topological features** and combined them with spectral and statistical features.
- Trained classification models; work submitted to **ICASSP 2026** and currently under review.

EDUCATION & TRAINING

01/08/2023 - Current - ROORKEE, INDIA

B.TECH. ELECTRONICS AND COMMUNICATIONS- INDIAN INSTITUTE OF TECHNOLOGY, ROORKEE

Relevant Courses: Fundamentals of AI/ML, Data Science, Biosignal & Image Analysis, Numerical Methods

Final grade: 8.744/10

SKILLS

Software & Programming

Python (ML ecosystem: PyTorch / TensorFlow) | High-performance computing tooling (e.g., MPI, CUDA, Slurm) | AlphaFold/ESMFold | ChimeraX | Dynamo,Relion | MATLAB

Methods & Approaches

Generative modeling (diffusion models) | Representation learning for biological sequences and structures | Causal language modeling (CLM) | Multi-domain learning of Language models | Fine-tuning with adapters (LoRA) | Fourier analysis | Topological Data Analysis | Physics-Informed Neural Network | Geometric Deep Learning | Curriculum Learning | Neuroevolutionary Networks

PROJECTS

01/02/2025 - 02/07/2025

Physics-informed neural network of heat-transfer for temperature field reconstruction Designed a Physics-Informed Neural Network to interpolate temperature fields from coarse (35) to fine (815) grids. Incorporated heat transfer PDE constraints into the loss function to ensure physically consistent predictions

02/07/2024 - 01/12/2024

MaskGAN for Shadow Removal: Conditional GAN model for shadow detection and removal

10/06/2024 - 02/12/2024

Model compression and benchmarking for edge devices Built a benchmark to evaluate ML models on edge devices across latency, accuracy, memory, and power metrics. Compressed VGG16 and ResNet50 to TensorFlow Lite using techniques like quantization , achieving up to 90% size reduction and ~10% power savings with only ~1% accuracy loss on an Intel i5 baseline.

HONOURS AND AWARDS

14/05/2025 Micron

Micron Memory Makers Hackathon — 1st Place (2025) Developed the top-performing and most lightweight model in the competition, optimized for edge deployment with a significantly reduced memory footprint

01/12/2022 Indian Association of Physics Teachers (IAPT)

NSEP State Topper — 2022 Among top performers in the National Standard Examination in Physics Olympiad

17/06/2023 JOSAA

All India Rank 1359, JEE Advanced 2023 Out of approximately **180,000+ candidates** who qualified for this exam, placing me within the **top ~0.8% of all test-takers** nationwide

14/10/2024 New Jersey Institute of Technology

1st place globally in Jersey CTF IV With InfoSecIITR, we secured 1st place in this Capture The Flag (CTF) competition globally

PUBLICATIONS

TOPO-ESA: Topology-Driven EEG Signal Analysis for Cognitive Workload Recognition

K. Gaurav, N. Sharma, R. Kumar, T. K. Reddy Bollu — Submitted to ICASSP 2026 (under peer review).

VOLUNTEERING

01/07/2024 - 01/09/2025 Roorkee

Core Member — InfoSecIITR Conducted cybersecurity workshops; authored CTF challenges (BackdoorCTF 2024, n00bCTF 2025)